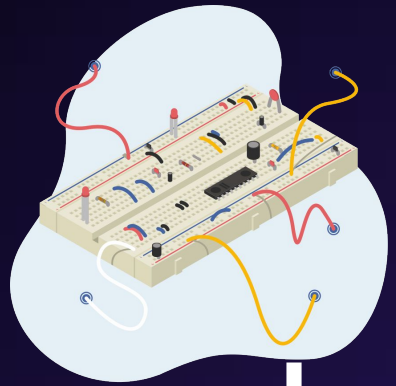
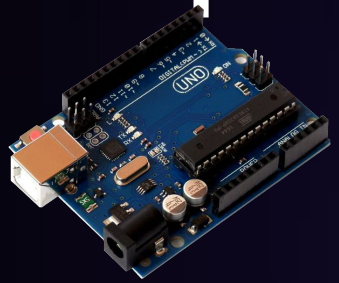
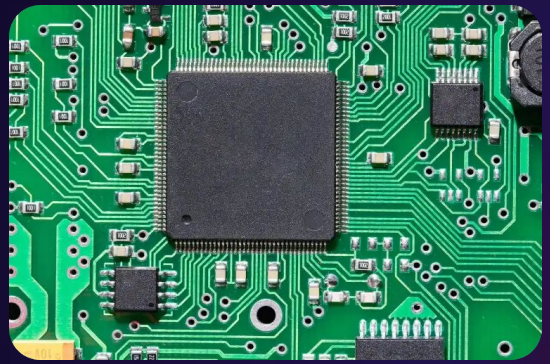
Architecture of electronic and computing machine and computing systems





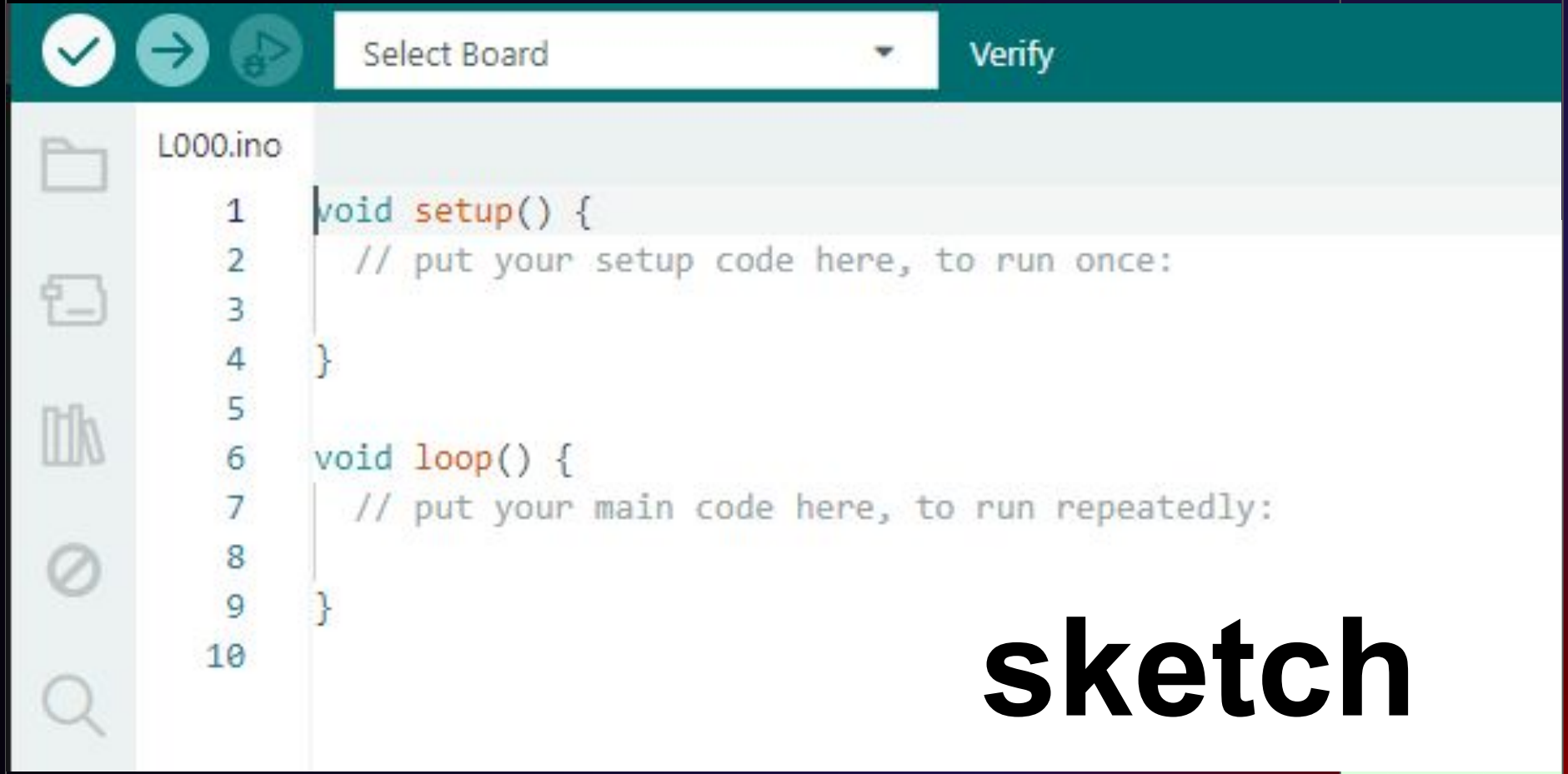


Course Code: S002
Author: Mirlan Nurbekov

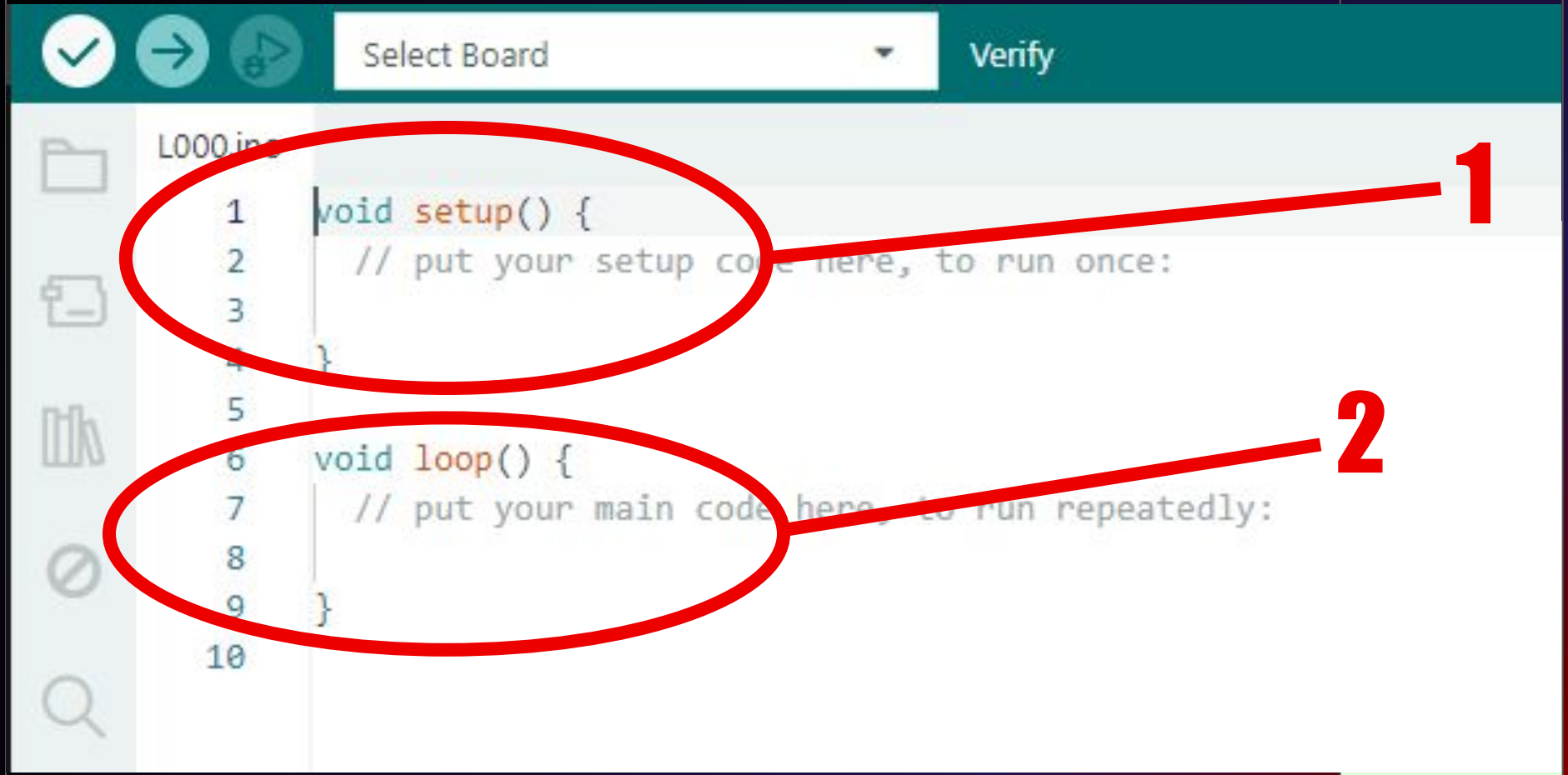
ALA-TOO INTERNATIONAL UNIVERSITY
IT & BUSINESS COLLEGE

What do you see when u open Arduino IDE




Select Board
Verify
L000.ino
1 void setup() {
2 // put your setup code here, to run once:
3
4 }
5
6 void loop() {
7 // put your main code here, to run repeatedly:
8
9 }
10

sketch


Select Board
Verify
1
2
1 void setup() {
2 // put your setup code here, to run once:
3
4 }
5
6 void loop() {
7 // put your main code here, to run repeatedly:
8
9 }
10

1

void means the function does NOT return any value

setup is the function name

( ) for parameters. It is empty cuz setup take NO input

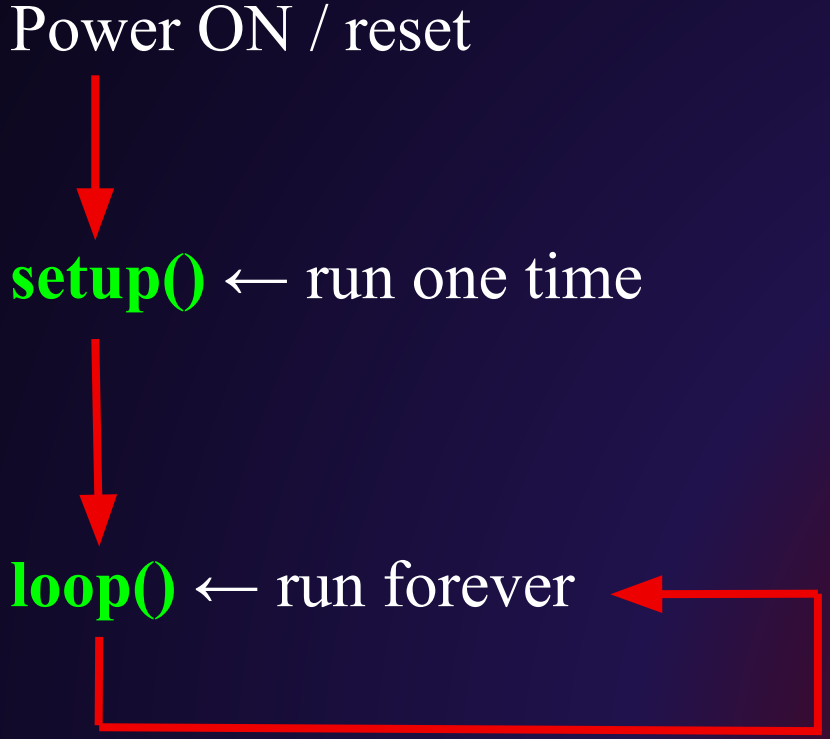
{ } is the body of function, code go in between brackets

2

loop code placed in the loop till system is off or if triggered by logic of the system. The hidden main is the reason for this result (basically it is an infinite for loop cuz no condition)

```
int main(void) {
  init();
  setup();
  for (;;) {        // the real infinite loop
    loop();         // your code runs once per round
  }
}
```

The program flow



Task L000

Write a program that makes the built-in LED turns ON for 1 second, OFF for 1 second and stops after 10 blinks.

Task L000

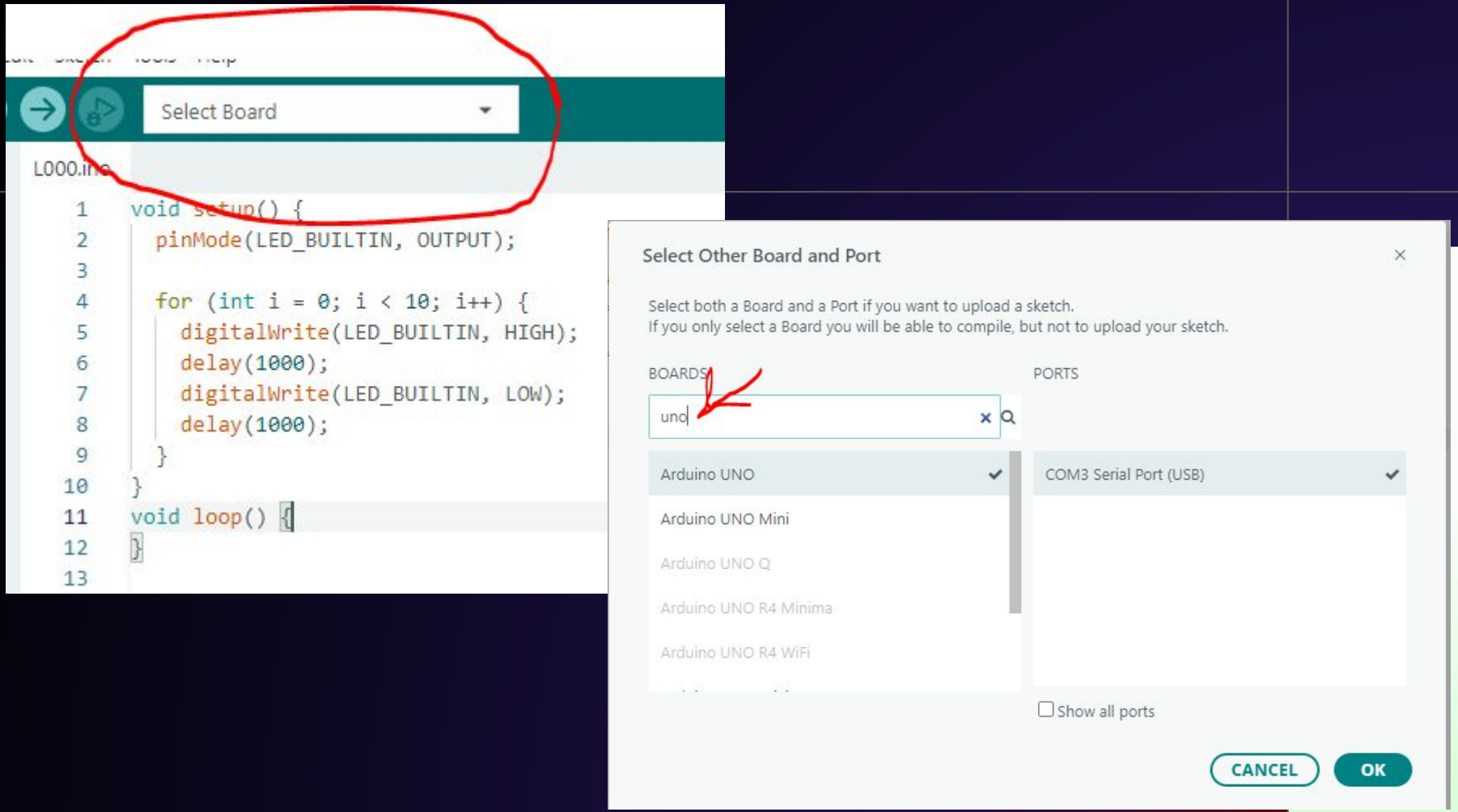
L000.ino

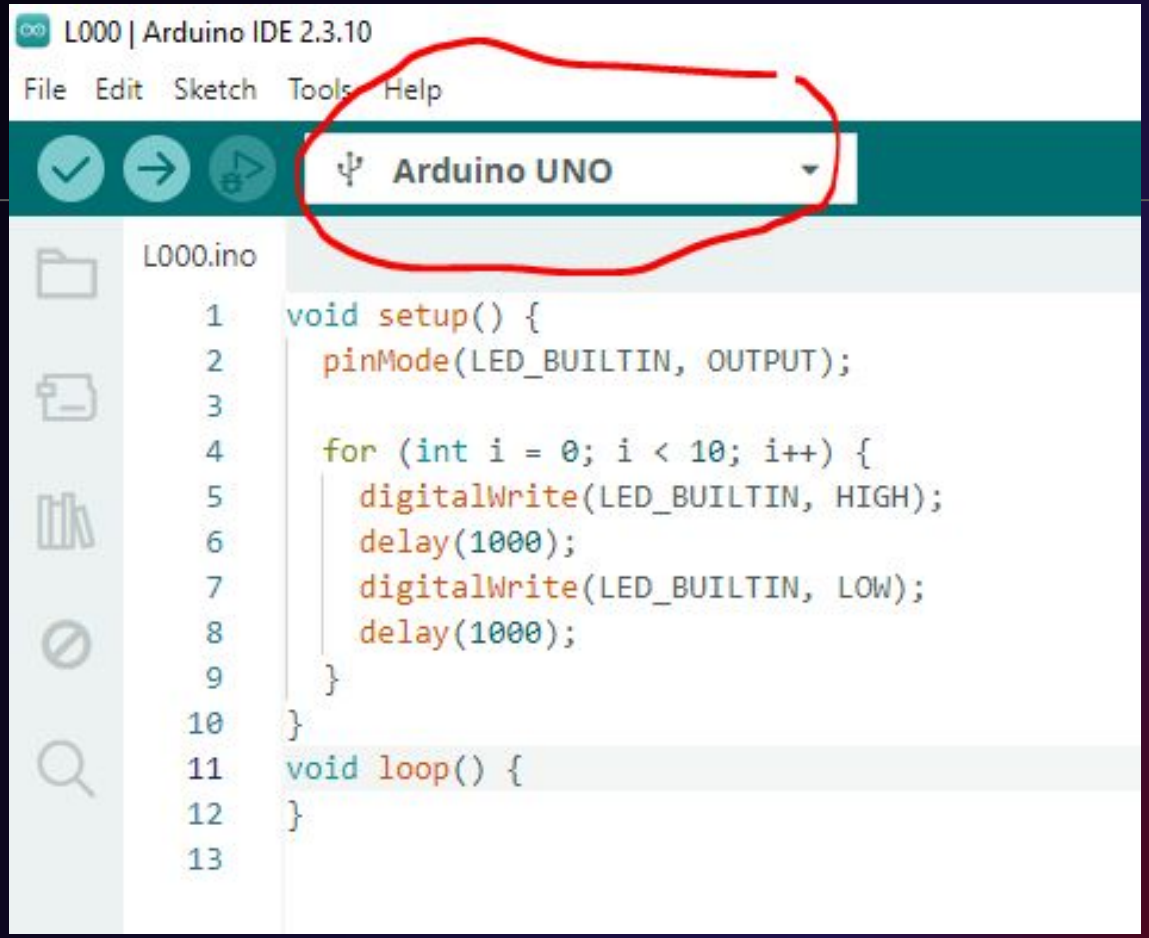
```
1   void setup() {
2     pinMode(LED_BUILTIN, OUTPUT);
3
4     for (int i = 0; i < 10; i++) {
5       digitalWrite(LED_BUILTIN, HIGH);
6       delay(1000);
7       digitalWrite(LED_BUILTIN, LOW);
8       delay(1000);
9     }
10  }
11  void loop() {
12  }
```

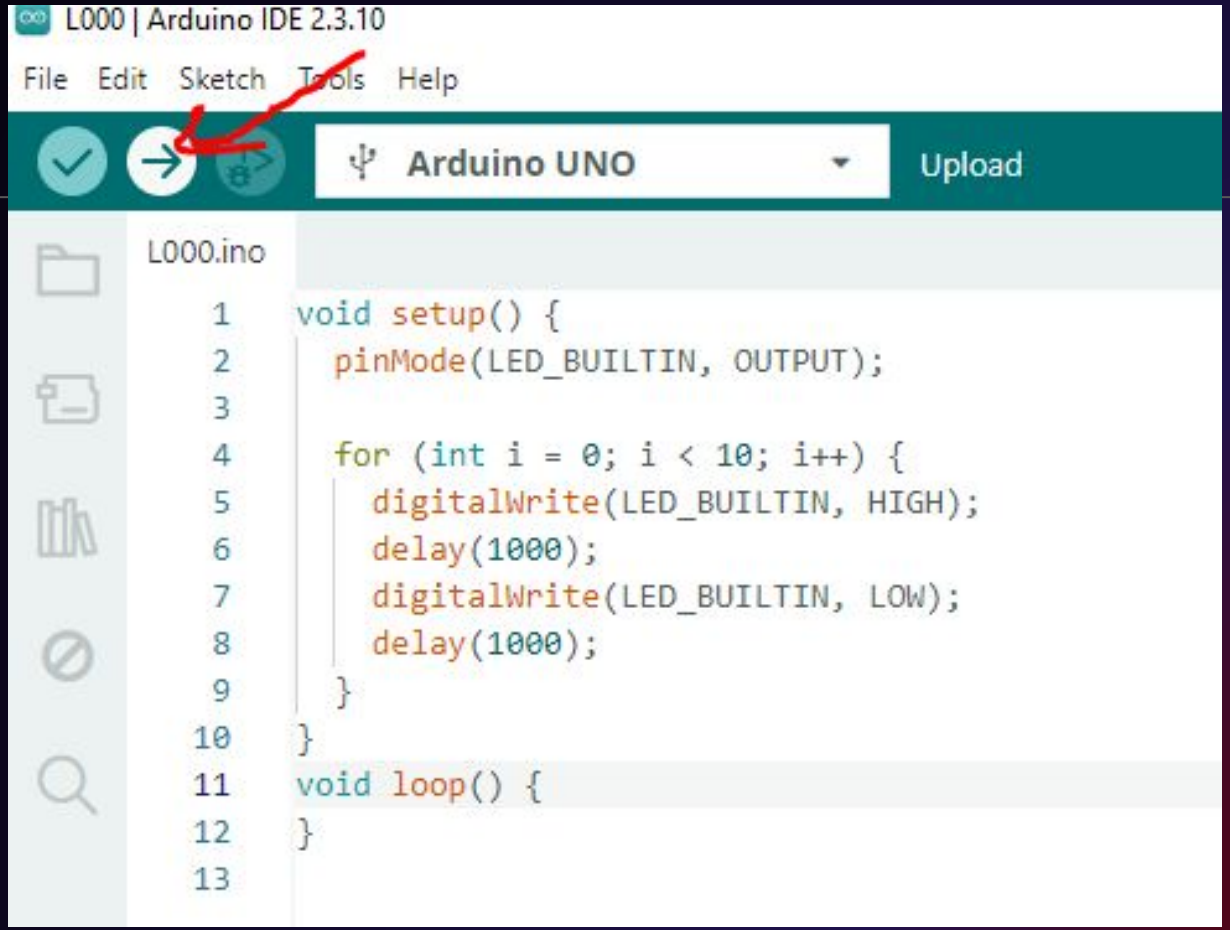
Task L000

L000.ino

```
1   void setup() {
2     pinMode(LED_BUILTIN, OUTPUT);       // configure the LED pin as output
3
4     for (int i = 0; i < 10; i++) {      // repeat 10 times
5       digitalWrite(LED_BUILTIN, HIGH);  // LED on
6       delay(1000);                      // wait 1 second
7       digitalWrite(LED_BUILTIN, LOW);   // LED off
8       delay(1000);                      // wait 1 second
9     }
10  }
11  void loop() {
12  }
13
```


Select Board
L000.ino
1 void setup() {
2 pinMode(LED_BUILTIN, OUTPUT);
3
4 for (int i = 0; i < 10; i++) {
5 digitalWrite(LED_BUILTIN, HIGH);
6 delay(1000);
7 digitalWrite(LED_BUILTIN, LOW);
8 delay(1000);
9 }
10 }
11 void loop() {
12 }
13
Select Other Board and Port
Select both a Board and a Port if you want to upload a sketch.
If you only select a Board you will be able to compile, but not to upload your sketch.
BOARDS
PORTS
uno
Arduino UNO
Arduino UNO Mini
Arduino UNO Q
Arduino UNO R4 Minima
Arduino UNO R4 WiFi
COM3 Serial Port (USB)
Show all ports
CANCEL
OK


L000 | Arduino IDE 2.3.10
File Edit Sketch Tools Help
Arduino UNO
L000.ino
1 void setup() {
2 pinMode(LED_BUILTIN, OUTPUT);
3
4 for (int i = 0; i < 10; i++) {
5 digitalWrite(LED_BUILTIN, HIGH);
6 delay(1000);
7 digitalWrite(LED_BUILTIN, LOW);
8 delay(1000);
9 }
10 }
11 void loop() {
12 }
13


L000 | Arduino IDE 2.3.10
File Edit Sketch Tools Help
Arduino UNO
Upload
L000.ino
1 void setup() {
2 pinMode(LED_BUILTIN, OUTPUT);
3
4 for (int i = 0; i < 10; i++) {
5 digitalWrite(LED_BUILTIN, HIGH);
6 delay(1000);
7 digitalWrite(LED_BUILTIN, LOW);
8 delay(1000);
9 }
10 }
11 void loop() {
12 }
13


Uploading...
CANCEL
Done compiling.

Can we use a global variable for T000



Task L000 (with global variable)

L000.ino

```
1  int count = 0;
2
3  void setup() {
4    pinMode(LED_BUILTIN, OUTPUT);
5  }
6
7  void loop() {
8    if (count < 10) {
9      digitalWrite(LED_BUILTIN, HIGH);
10     delay(1000);
11     digitalWrite(LED_BUILTIN, LOW);
12     delay(1000);
13     count++;
14   }
15 }
16
```

Task L000 (with global variable)

L000.ino

```
1   int count = 0;   // global variable: keeps its value between loop runs
2
3   void setup() {
4     pinMode(LED_BUILTIN, OUTPUT);
5   }
6
7   void loop() {
8     if (count < 10) {
9       digitalWrite(LED_BUILTIN, HIGH);
10      delay(1000);
11      digitalWrite(LED_BUILTIN, LOW);
12      delay(1000);
13      count++;       // one more blink done
14    }
15  }
```

Is a global variable good use for T000



Is a global variable good use for T000

NO

Task L001

Write a program that makes the built-in LED turns ON for 500 ms, OFF for 2000 ms and stops after 10 blinks

Task L001

L001.ino

```
1   void setup() {
2     pinMode(LED_BUILTIN, OUTPUT);
3
4     for (int i = 0; i < 10; i++) {
5       digitalWrite(LED_BUILTIN, HIGH);
6       delay(500);
7       digitalWrite(LED_BUILTIN, LOW);
8       delay(2000);
9     }
10  }
11
12  void loop() {
13  }
```

Task L001

L001.ino

```
1   void setup() {
2     pinMode(LED_BUILTIN, OUTPUT);
3
4     for (int i = 0; i < 10; i++) {
5       digitalWrite(LED_BUILTIN, HIGH);
6       delay(500);                          // ON for 500 ms
7       digitalWrite(LED_BUILTIN, LOW);
8       delay(2000);                         // OFF for 2000 ms
9     }
10  }
11
12  void loop() {
13  }
```

Task L002

Write a program that makes the built-in LED turns ON for 1000 ms, OFF for 1000 ms and stops after 5 blinks

Task L002

L002.ino

```
1   void setup() {
2     pinMode(LED_BUILTIN, OUTPUT);
3
4     for (int i = 0; i < 5; i++) {
5       digitalWrite(LED_BUILTIN, HIGH);
6       delay(1000);
7       digitalWrite(LED_BUILTIN, LOW);
8       delay(1000);
9     }
10  }
11
12  void loop() {
13  }
```

Task L002

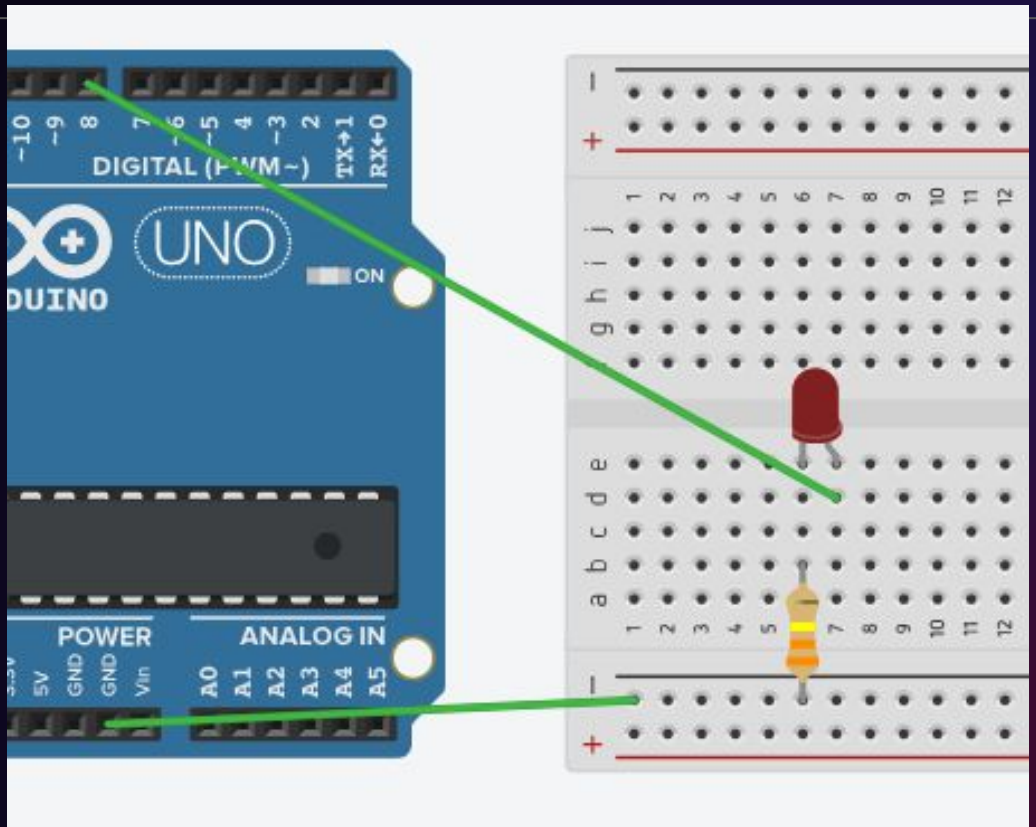
L002.ino

```
1   void setup() {
2     pinMode(LED_BUILTIN, OUTPUT);
3
4     for (int i = 0; i < 5; i++) {        // number of blinks
5       digitalWrite(LED_BUILTIN, HIGH);
6       delay(1000);
7       digitalWrite(LED_BUILTIN, LOW);
8       delay(1000);
9     }
10  }
11
12  void loop() {
13  }
```

Task L003

Build the circuit on a breadboard and connect it to the Uno. Write a program so the LED blinks 3 times: ON for 3000 ms, then OFF for 3000 ms each time. After the third blink the LED stays off.

Task L003


DIGITAL (PWM~)
TX→1
RX←0
UNO
ON
POWER
ANALOG IN
5V
GND
GND
Vin
A0
A1
A2
A3
A4
A5

Task L003

L003.ino

```
1   const int LED_PIN = 8;                  // LED is connected to pin 8
2
3   void setup() {
4     pinMode(LED_PIN, OUTPUT);             // pin 8 sends the signal out
5
6     for (int i = 0; i < 3; i++) {         // repeat 3 times
7       digitalWrite(LED_PIN, HIGH);        // LED on
8       delay(3000);                        // wait 3 seconds
9       digitalWrite(LED_PIN, LOW);         // LED off
10      delay(3000);                        // wait 3 seconds
11    }
12  }
13
14  void loop() {
15  }
```

Task L004

Use circuit from L003 and write a program so the LED blinks 100 times: ON for 50 ms, then OFF for 50 ms each time.

Task L004

L004.ino

```
1   const int LED_PIN = 8;
2   void setup() {
3     pinMode(LED_PIN, OUTPUT);
4
5     for (int i = 0; i < 100; i++) {
6       digitalWrite(LED_PIN, HIGH);
7       delay(50);
8       digitalWrite(LED_PIN, LOW);
9       delay(50);
10    }
11  }
12
13  void loop() {
14  }
```

Task L005

Use the circuit from L003. Write a program that sends the Morse code signal SOS with the LED:

S = 3 short flashes (200 ms ON, 200 ms OFF)
O = 3 long flashes (600 ms ON, 200 ms OFF)
S = 3 short flashes again

Then a pause of 2000 ms and the signal repeats forever.

Task L005

L005.ino

```
1   const int LED_PIN = 8;
2   void flash(int onTime) {
3     digitalWrite(LED_PIN, HIGH);
4     delay(onTime);
5     digitalWrite(LED_PIN, LOW);
6     delay(200);
7   }
8   void setup() {
9     pinMode(LED_PIN, OUTPUT);
10  }
11  void loop() {
12    for (int i = 0; i < 3; i++) flash(200);
13    for (int i = 0; i < 3; i++) flash(600);
14    for (int i = 0; i < 3; i++) flash(200);
15    delay(2000);
16  }
```

SMALL GAME TIME



Task L006

Build a circuit with two LEDs and write a program so they take turns. LED 1 is ON for 1000 ms while LED 2 is OFF. Then LED 1 is OFF for 1000 ms while LED 2 is ON. Each LED must blink 10 times. After that both LEDs stay off. The LEDs must never be on at the same time, and the whole program runs for 20 seconds.

Task L006

L006.ino

```ino
const int LED1_PIN = 8;              // first LED
const int LED2_PIN = 9;              // second LED

void setup() {
  pinMode(LED1_PIN, OUTPUT);
  pinMode(LED2_PIN, OUTPUT);

  for (int i = 0; i < 10; i++) {     // repeat 10 times
    digitalWrite(LED1_PIN, HIGH);    // LED 1 on
    digitalWrite(LED2_PIN, LOW);     // LED 2 off
    delay(1000);

    digitalWrite(LED1_PIN, LOW);     // LED 1 off
    digitalWrite(LED2_PIN, HIGH);    // LED 2 on
    delay(1000);
  }

  digitalWrite(LED2_PIN, LOW);       // LED 2 would stay on after the last round
}

void loop() {
}
```

Task L007

Build a circuit with three LEDs and write a program so they take turns. Each LED is ON for 500 ms, and only one LED can be on at a time. The order is LED 1, LED 2, LED 3, then LED 1 again, and so on. The program must never stop.

Task L007

L007.ino

```
1  const int LED1_PIN = 8;
2  const int LED2_PIN = 9;
3  const int LED3_PIN = 10;
4
5  void setup() {
6    pinMode(LED1_PIN, OUTPUT);
7    pinMode(LED2_PIN, OUTPUT);
8    pinMode(LED3_PIN, OUTPUT);
9  }
10
11 void loop() {
12   digitalWrite(LED1_PIN, HIGH);
13   delay(500);
14   digitalWrite(LED1_PIN, LOW);
15
16   digitalWrite(LED2_PIN, HIGH);
17   delay(500);
18   digitalWrite(LED2_PIN, LOW);
19
20   digitalWrite(LED3_PIN, HIGH);
21   delay(500);
22   digitalWrite(LED3_PIN, LOW);
23 }
```

Task L008

Build a circuit with four LEDs and write a program that counts from 0 to 10 in binary. Each number is shown for 2000 ms. ON LED means 1, an LED that is OFF means 0. After 10 the counter goes back to 0 and the program never stops.

Decimal	Binary
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
10	1010

Task L008

L008.ino

```ino
const int LED1_PIN = 8;
const int LED2_PIN = 9;
const int LED3_PIN = 10;
const int LED4_PIN = 11;

void setup() {
  pinMode(LED1_PIN, OUTPUT);
  pinMode(LED2_PIN, OUTPUT);
  pinMode(LED3_PIN, OUTPUT);
  pinMode(LED4_PIN, OUTPUT);
}

void loop() {
  for (int number = 0; number <= 10; number++) {
    digitalWrite(LED1_PIN, bitRead(number, 3));
    digitalWrite(LED2_PIN, bitRead(number, 2));
    digitalWrite(LED3_PIN, bitRead(number, 1));
    digitalWrite(LED4_PIN, bitRead(number, 0));
    delay(2000);
  }
}
```

Self test mcq

1. What does **void** mean in **void setup()** ?

A) The function accepts no input

B) The function runs forever

C) The function returns nothing

D) The function has no name

2. When does the code inside **setup()** run?

A) Again and again until power is switched off

B) Once, right after power on or reset

C) Only after a button is pressed

D) Once after each round of loop()

Self test mcq

3. In the hidden **main()**, how is **loop()** executed?

A) Runs it once, right after setup()

B) Runs it only when an input changes

C) Runs it before setup(), every time

D) Calls it again and again, forever

Self test mcq

4. In the global-variable version of L000, why is int count = 0; declared outside setup() and loop()?

A) To keep its value between loop() runs

B) So the LED can blink much faster

C) So pinMode() can be skipped later on

D) So delay() can wait longer than 1000 ms

Self test mcq

5. In L008, bitRead(number, 3) controls LED 1 and bitRead(number, 0) controls LED 4. Which LEDs are ON when the number 5 is shown?

A) LEDs 1 and 3

B) LEDs 1 and 2

C) LEDs 3 and 4

D) LEDs 2 and 4

Answers

1. C
2. B
3. D
4. A
5. D

Home work

=1= Recreate all circuits from today's lab session in Tinkercad Circuits and simulate them with your code.

=2= Choose one component you like (buzzer, push button, potentiometer, servo, RGB LED, ultrasonic sensor...). Read about how it works and try it in the simulator.

Be ready to show and explain it at the next class.